

Cisatracurium: Drug information - UpToDate

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Cisatracurium: Drug information

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For additional information see "Cisatracurium: Patient drug information" and "Cisatracurium: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Nimbex [DSC]

Pharmacologic Category

- Neuromuscular Blocker Agent, Nondepolarizing

Dosing: Adult

Dosage guidance :

Dosing: Doses must be individualized due to interpatient variability, patient-specific factors, and/or comorbidities (eg, severe trauma, burn injury) (Ref).

Endotracheal intubation

Note: Ensure adequate pain control and sedation prior to and during administration of neuromuscular blockade.

Endotracheal intubation: IV: 0.15 to 0.2 **mg/kg** once.

Rapid sequence intubation, neuromuscular blockade (alternative agent) (off-label use):

Note: Experts only consider use when succinylcholine or rocuronium is contraindicated or not available; may have a delayed onset of action when compared to succinylcholine or rocuronium (Ref).

IV: 0.4 **mg/kg** once (Ref).

Mechanically ventilated patients in the ICU, neuromuscular blockade

Mechanically ventilated patients in the ICU, neuromuscular blockade: Note: Ensure adequate pain control and sedation prior to and during administration of neuromuscular blockade. May use to facilitate mechanical ventilation (eg, refractory moderate to severe acute respiratory distress syndrome; refractory, life-threatening status asthmaticus; shivering from therapeutic hypothermia) (Ref). Refer to institutional protocols; titration specifics may vary.

Continuous infusion: IV: 0.1 to 0.2 **mg/kg** initial loading dose, followed by continuous infusion of 1 to 3 **mcg/kg/minute**; titrate based on clinical response and neuromuscular monitoring (eg, train-of-four); usual dosage range: 0.5 to 10 **mcg/kg/minute** (Ref).

Neuromuscular blockade during general anesthesia

Neuromuscular blockade during general anesthesia (adjunctive therapy):

Maintenance dosing: Note: Inhaled anesthetic agents (eg, desflurane, isoflurane, sevoflurane) prolong the duration of action of cisatracurium and may potentiate the neuromuscular blockade.

Intermittent dose: IV: 0.01 to 0.03 **mg/kg**, may repeat as needed based on clinical response and neuromuscular monitoring (eg, train-of-four) (Ref).

Continuous infusion: IV: 1 to 3 **mcg/kg/minute** as an initial continuous infusion; titrate as needed based on clinical response and neuromuscular monitoring (eg, train-of-four) (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

IV: No dosage adjustment necessary (Ref).

Note: The time to 90% neuromuscular block is ~1 minute slower in patients with end-stage kidney disease compared to those with normal kidney function. Consider extending the interval between administering cisatracurium and attempting intubation by at least 1 minute (Ref).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. The half-life values of the metabolites are longer in patients with hepatic disease and concentrations may be higher after long-term administration.

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Class 1, 2 or 3 obesity (BMI ≥ 30 kg/m²):

IV: Use **ideal body weight** for weight-based dosing calculations, then titrate to clinical effect (Ref). In patients with extreme obesity (eg, BMI ≥ 50 kg/m²), or when underdosing is a concern, consider using **adjusted body weight** for weight-based dosing calculations, then titrate to clinical effect (Ref). Clinicians should **not** change dosing weight from one weight metric to another during therapy (ie, actual body weight to/from either adjusted body weight or ideal body weight) (Ref). Refer to adult dosing for indication-specific doses.

Rationale for recommendations: Nondepolarizing neuromuscular blocking agents (NMBAs) are hydrophilic compounds with a small V_d , thus distribution of cisatracurium into adipose tissue is limited (Ref). There are no studies evaluating the most appropriate weight metric in patients with obesity receiving sustained continuous infusions. Data with NMBAs in patients with obesity predominantly originate from studies evaluating either single bolus or incremental doses in patients undergoing surgical procedures. One study in patients undergoing bariatric laparoscopic gastric banding reported an ~26% prolonged duration of effect when cisatracurium dosing was based on actual body weight compared to ideal body weight (Ref). Another study suggested similar onset times but shorter recovery durations (~19% shorter) in patients with extreme obesity (mean BMI = 55 kg/m²) when cisatracurium was dosed using adjusted body weight versus a control group of patients without obesity (Ref).

Dosing: Older Adult

Refer to adult dosing. Consider extending the interval between administration of cisatracurium and intubation attempt by at least 1 minute to achieve adequate intubation conditions (time to maximum neuromuscular blockade is ~1 minute slower in elderly patients).

Dosing: Pediatric

(For additional information see "Cisatracurium: Pediatric drug information")

Note: Dosing in obese patients should be calculated using ideal body weight (Ref):

Neuromuscular blockade

Neuromuscular blockade:

Intermittent dosing:

Infants and Children <2 years: IV: 0.15 mg/kg/dose over 5 to 10 seconds.

Children ≥2 years and Adolescents: IV: 0.1 to 0.15 mg/kg/dose over 5 to 10 seconds (Ref).

Continuous IV infusion: Infants, Children, and Adolescents: 1 to 4 **mcg**/kg/minute (0.06 to 0.24 mg/kg/**hour**); 3.9 ± 1.3 **mcg**/kg/minute (0.23 ± 0.08 mg/kg/**hour**) was the mean infusion rate needed to maintain neuromuscular blockade in 19 children (ages 3 months to 16 years) (Ref); higher doses have been reported with prolonged infusions (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling. Slower times to onset of complete neuromuscular block were observed in renal dysfunction patients; extending the interval between the administration of cisatracurium and intubation attempt may be required to achieve adequate intubation conditions.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling. The time to onset of action was ~1 minute faster in liver transplant patients but was not associated with clinically significant changes in recovery time.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Reported incidences are for cisatracurium in conjunction with other drugs.

1% to 10%: Respiratory: Bronchospasm (≤1%)

<1%:

Cardiovascular: Bradycardia, flushing, hypotension

Dermatologic: Skin rash

Postmarketing:

Hypersensitivity: Histamine release, severe hypersensitivity reaction (including anaphylaxis)

Nervous system: Delayed recovery from neuromuscular block, myasthenia

Neuromuscular & skeletal: Myopathy

Miscellaneous: Malignant hyperthermia (Zelege 2024)

Contraindications

Hypersensitivity to cisatracurium or any component of the formulation; use of the 10 mL multiple-dose vials in neonates <1 month of age and low birth-weight infants (formulation contains benzyl alcohol).

Warnings/Precautions

Concern related to adverse effects:

- Anaphylaxis: Severe anaphylactic reactions have been reported (some life-threatening and fatal). Appropriate emergency treatment (including epinephrine 1 mg/mL) should be immediately available during use. Use caution in patients with previous anaphylactic reactions to other neuromuscular blocking agents.

- Bradycardia: May be more common with cisatracurium than with other neuromuscular-blocking agents since it has no clinically-significant effects on heart rate to counteract the bradycardia produced by anesthetics.
- Residual paralysis: Has been reported. Patients with neuromuscular diseases (eg, myasthenia gravis, myasthenic syndrome) and carcinomatosis may be at increased risk; a lower maximum initial bolus dose is recommended in these patients. Reversal agents may be considered, especially in cases where residual paralysis is more likely to occur. To prevent complications from residual paralysis, extubation is recommended only after the patient has sufficiently recovered from neuromuscular blockade.

Disease-related concerns:

- Burn injury: Resistance may occur in burn patients ($\geq 20\%$ of total body surface area), usually several days after the injury, and may persist for several months after wound healing (Han 2009).
- Conditions that may *antagonize* neuromuscular blockade (decreased paralysis): Respiratory alkalosis, hypercalcemia, demyelinating lesions, peripheral neuropathies, denervation, and muscle trauma may result in antagonism of neuromuscular blockade (ACCM/SCCM/ASHP [Murray 2002]; Greenberg 2013; Miller 2010; Naguib 2002).
- Conditions that may *potentiate* neuromuscular blockade (increased paralysis): Electrolyte abnormalities (eg, severe hypocalcemia, severe hypokalemia, hypermagnesemia), neuromuscular diseases, metabolic acidosis, respiratory acidosis, Eaton-Lambert syndrome and myasthenia gravis may result in potentiation of neuromuscular blockade (Greenberg 2013; Miller 2010; Naguib 2002).
- Therapeutic hypothermia: Hypothermia may slow Hoffmann elimination thereby prolonging the duration of paralysis (Greenberg 2013).

Special populations:

- Older adult: Use with caution in the elderly, effects and duration are more variable.
- Immobilized patients: Resistance may occur in patients who are immobilized.

Dosage form specific issues:

- Benzyl alcohol and derivatives: Some dosage forms may contain benzyl alcohol; large amounts of benzyl alcohol (≥ 99 mg/kg/day) have been associated with a potentially fatal toxicity (“gasping syndrome”) in neonates; the “gasping syndrome” consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension and cardiovascular collapse (AAP [“Inactive” 1997]; CDC, 1982); some data suggests that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol with caution in neonates. See manufacturer’s labeling.

Other warnings/precautions:

- Appropriate use: Maintenance of an adequate airway and respiratory support is critical. All patients should receive eye care including liberal use of lubricating drops, gel, or ointment and eyelids should remain closed during continuous neuromuscular blockade to protect against damage to the cornea (ulceration and drying).
- Experienced personnel: Should be administered by adequately trained individuals familiar with its use.
- Risk of medication errors: Accidental administration may be fatal. Confirm proper selection of intended product, store vial so the cap and ferrule are intact and the possibility of selecting the wrong product is minimized, and ensure that the intended dose is clearly labeled and communicated, when applicable.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Intravenous, as besylate:

Nimbex: 20 mg/10 mL (10 mL [DSC]) [contains benzyl alcohol]

Generic: 10 mg/5 mL (5 mL [DSC]); 20 mg/10 mL (10 mL); 200 mg/20 mL (20 mL [DSC])

Solution, Intravenous, as besylate [preservative free]:

Nimbex: 10 mg/5 mL (5 mL [DSC]); 200 mg/20 mL (20 mL [DSC])

Generic: 10 mg/5 mL (5 mL); 200 mg/20 mL (20 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Cisatracurium Besylate (PF) Intravenous)

10 mg/5 mL (per mL): \$1.43 - \$3.61

200 mg/20 mL (per mL): \$2.67 - \$17.08

Solution (Cisatracurium Besylate Intravenous)

20 mg/10 mL (per mL): \$1.08 - \$3.16

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous, as besylate:

Generic: 20 mg/10 mL (10 mL)

Administration: Adult

IV: Administer IV only; do **not** administer IM (excessive tissue irritation). Administer undiluted as bolus injection over 5 to 10 seconds. Continuous infusion administration requires the use of an infusion pump. Do not administer with alkaline solutions simultaneously in the same IV line.

Preparation for Administration: Adult

May administer IV without dilution or may dilute to a concentration of 0.1 mg/mL in a compatible infusion solution (D5W, NS, D5NS); may also dilute to a concentration of 0.1 or 0.2 mg/mL in D5LR. Do not dilute in LR or alkaline solutions.

Usual Infusion Concentrations: Adult

IV infusion: 100 mg in 250 mL (total volume) (concentration: 400 **mcg**/mL) of D₅W or NS

Administration: Pediatric

Parenteral: Administer IV only; not for IM injection due to tissue irritation.

Rapid IV injection: Administer undiluted over 5 to 10 seconds.

Continuous IV infusion: Administer via infusion pump.

Preparation for Administration: Pediatric

Parenteral: Continuous IV infusion: Dilute in a compatible solution (eg, D5W, NS) for IV infusion; see product labeling for details. The manufacturer suggests a final concentration range of 0.1 to 0.2 mg/mL; however, final concentrations of 1 mg/mL, 2 mg/mL, and 5 mg/mL have been used by some centers (Ref). For pediatric patients, ASHP recommends standard concentrations of 1 mg/mL and 2 mg/mL (undiluted) (Ref). Some dosage forms may contain benzyl alcohol; see product-specific label for details; in neonates, consider using single-use vials for preparation to avoid benzyl alcohol.

Storage/Stability

Refrigerate intact vials at 2°C to 8°C (36°F to 46°F). Use vials within 21 days upon removal from the refrigerator to room temperature of 25°C (77°F). Per the manufacturer, dilutions of 0.1 mg/mL in 0.9% sodium chloride (NS), dextrose 5% in water (D₅W), or D₅NS are stable for up to 24 hours at room temperature or under refrigeration; dilutions of 0.1 or 0.2 mg/mL in D₅LR are stable for up to 24 hours in the refrigerator. *Additional stability data:* Dilutions of 0.1, 2, and 5 mg/mL in D₅W or NS are stable in the refrigerator for up to 30 days; at room temperature (23°C), dilutions of 0.1 and 2 mg/mL began exhibiting substantial drug loss between 7-14 days; dilutions of 5 mg/mL in D₅W or NS are stable for up to 30 days at room temperature (23°C) (Xu 1998). Usual concentration: 0.1 to 0.4 mg/mL.

According to the 2020-2021 ISMP Targeted Medication Safety Best Practices for Hospitals, neuromuscular blockers should be segregated, sequestered, and differentiated from all other medication wherever they are stored (ISMP 2020).

Use: Labeled Indications

Endotracheal intubation: To facilitate endotracheal intubation.

Mechanically ventilated patients in the ICU, neuromuscular blockade: Provide skeletal muscle relaxation during mechanical ventilation in ICU patients.

Neuromuscular blockade during general anesthesia: As an adjunct to general anesthesia to provide skeletal muscle relaxation during surgery.

Note: Neuromuscular blockade does not provide pain control, sedation, or amnestic effects. Ensure adequate pain control and sedation prior to and during administration of neuromuscular blockade (SCCM [Murray 2016]).

Limitations of use: Cisatracurium is not recommended for rapid sequence endotracheal intubation due to the time required for its onset of action, but may be considered when first-line options (eg, rocuronium, succinylcholine) are contraindicated or unavailable.

Use: Off-Label: Adult

Rapid sequence intubation, neuromuscular blockade

Medication Safety Issues

Sound-alike/look-alike issues:

Nimbex may be confused with NovoLOG

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (neuromuscular blocker agent) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Other safety concerns:

According to the 2020-2021 ISMP Targeted Medication Safety Best Practices for Hospitals, neuromuscular blockers should be segregated, sequestered, and differentiated from all other medication wherever they are stored. This includes:

Only storing in places within the hospital that they are routinely used.

Placing in sealed boxes or in rapid sequence intubation kits (preferred).

Limiting availability in automated dispensing cabinets to perioperative, labor and delivery, critical care, and emergency departments only.

Placing in separate lidded containers within the pharmacy refrigerator or other isolated pharmacy storage area.

Affixing an auxiliary label to clearly communicate respiratory paralysis will occur and ventilation required on all storage bins and/or automated dispensing pockets/drawers (exception anesthesia-prepared syringes) stating one of the following:

Warning: Causes Respiratory Arrest - Patient Must Be Ventilated

Warning: Paralyzing Agent - Causes Respiratory Arrest

Warning: Causes Respiratory Paralysis - Patient Must Be Ventilated

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Acetylcholinesterase Inhibitors: May decrease neuromuscular-blocking effects of Neuromuscular-Blocking Agents (Nondepolarizing). *Risk C: Monitor*

Aminoglycosides: May increase therapeutic effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Bacitracin (Systemic): May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Botulinum Toxin-Containing Products: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Bromperidol: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Calcium Channel Blockers: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents (Nondepolarizing). *Risk C: Monitor*

Capreomycin: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Carbamazepine: May decrease serum concentrations of Neuromuscular-Blocking Agents (Nondepolarizing). *Risk C: Monitor*

Cardiac Glycosides: Neuromuscular-Blocking Agents may increase arrhythmogenic effects of Cardiac Glycosides. *Risk C: Monitor*

Clindamycin (Topical): May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Colistimethate: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. Management: If possible, avoid concomitant use of these products. Monitor for deeper, pro-

longed neuromuscular-blocking effects (respiratory paralysis) in patients receiving concomitant neuromuscular-blocking agents and colistimethate. *Risk D: Consider Therapy Modification*

Corticosteroids (Systemic): Neuromuscular-Blocking Agents (Nondepolarizing) may increase adverse neuromuscular effects of Corticosteroids (Systemic). Increased muscle weakness, possibly progressing to polyneuropathies and myopathies, may occur. Management: If concomitant therapy is required, use the lowest dose for the shortest duration to limit the risk of myopathy or neuropathy. Monitor for new onset or worsening muscle weakness, reduction or loss of deep tendon reflexes, and peripheral sensory decrements *Risk D: Consider Therapy Modification*

CycloSPORINE (Systemic): May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Esketamine (Injection): May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Fosphenytoin-Phenytoin: May decrease neuromuscular-blocking effects of Neuromuscular-Blocking Agents (Nondepolarizing). Specifically, this effect is observed with chronic phenytoin administration. Fosphenytoin-Phenytoin may increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents (Nondepolarizing). Specifically, this effects is observed with acute/short term phenytoin administration. *Risk C: Monitor*

Gepotidacin: May decrease therapeutic effects of Neuromuscular-Blocking Agents (Nondepolarizing). *Risk C: Monitor*

Inhalational Anesthetics: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents (Nondepolarizing). Management: When initiating a non-depolarizing neuromuscular blocking agent (NMBA) in a patient receiving an inhalational anesthetic, initial NMBA doses should be reduced 15% to 25% and doses of continuous infusions should be reduced 30% to 60%. *Risk D: Consider Therapy Modification*

Ketamine: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Ketorolac (Nasal): May increase adverse/toxic effects of Neuromuscular-Blocking Agents (Nondepolarizing). Specifically, episodes of apnea have been reported in patients using this combination. *Risk C: Monitor*

Ketorolac (Systemic): May increase adverse/toxic effects of Neuromuscular-Blocking Agents (Nondepolarizing). Specifically, episodes of apnea have been reported in patients using this combination. *Risk C: Monitor*

Lincosamide Antibiotics: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Lithium: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Local Anesthetics: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Loop Diuretics: May decrease neuromuscular-blocking effects of Neuromuscular-Blocking Agents. Loop Diuretics may increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Magnesium Salts: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Minocycline (Systemic): May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Pancuronium: May increase adverse/toxic effects of Neuromuscular-Blocking Agents. Pancuronium may increase therapeutic effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Pholcodine: May increase adverse/toxic effects of Neuromuscular-Blocking Agents. Specifically, anaphylaxis has been reported. *Risk C: Monitor*

Polymyxin B: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Procainamide: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

QuiNIDine: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

QuiNINE: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk X: Avoid*

Tetracyclines: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Thiazide and Thiazide-Like Diuretics: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents (Nondepolarizing). *Risk C: Monitor*

Vancomycin: May increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Pregnancy Considerations

Following administration of atracurium, laudanosine, an active metabolite of atracurium and cisatracurium, crosses the placenta (Atherton 1999; Fodale 2002).

Recommendations for using cisatracurium for sustained neuromuscular blockade in pregnant critically ill patients are not available due to insufficient evidence (SCCM [Murray 2016]).

Use with caution in patients treated for preeclampsia/eclampsia; the action of cisatracurium may be enhanced by concomitant medications (consult drug interactions database). Preparations may contain benzyl alcohol; avoid in pregnant patients due to association with gasping syndrome in premature infants.

Breastfeeding Considerations

It is not known if cisatracurium is present in breast milk.

In general, neuromuscular blocking agents may be used in lactating patients based on their physical characteristics and poor oral bioavailability (ABM [Reece-Stremtan 2017]). Preparations may contain benzyl alcohol; avoid in lactating patients due to association with gasping syndrome in premature infants. According to the manufacturer, the decision to breastfeed should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother.

Monitoring Parameters

Vital signs (heart rate, blood pressure, respiratory rate); degree of muscle paralysis (eg, presence of spontaneous movement, ventilator asynchrony, shivering, and use of a peripheral nerve stimulator with train of four monitoring along with clinical assessments)

In the ICU setting, limiting the duration of paralysis and providing a physiotherapy regimen in patients requiring continuous paralysis is suggested.

Mechanism of Action

Blocks neural transmission at the myoneural junction by binding with cholinergic receptor sites

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: IV: 2 to 3 minutes (dose dependent).

Peak effect: 3 to 5 minutes.

Duration: Dose dependent, 35 to 45 minutes after a single 0.1 mg/kg dose; recovery begins in 20 to 35 minutes when anesthesia is balanced; recovery is attained in 90% of patients in 25 to 93 minutes; hypothermia may prolong the duration of action (NCS [Madden 2017]).

Distribution: V_{dss} : 145 mL/kg (21% larger V_{dss} when receiving inhalational anesthetics).

Protein binding: Not studied due to rapid degradation at physiologic pH.

Metabolism: Undergoes rapid nonenzymatic degradation in the bloodstream (Hofmann elimination) to laudanosine and inactive metabolites; laudanosine may cause CNS stimulation (association not established in humans) and has less accumulation with prolonged use than atracurium due to lower requirements for clinical effect.

Half-life elimination: 22 to 29 minutes.

Excretion: Urine (95%; <10% as unchanged drug); feces (4%).

Clearance: Children: 5.89 mL/kg/minute; Adults: 4.57 mL/kg/minute.

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Altered kidney function: The times to 90% block were approximately 1 minute slower in patients with ESRD. The half-life values of the metabolites are longer in patients with renal failure and concentrations may be higher after long-term administration.

Hepatic function impairment: The times to maximum block were approximately 1 minute faster in liver transplant recipients compared with healthy adults. The half-life values of the metabolites are longer in patients with hepatic disease and concentrations may be higher after long-term administration.

Pediatric: Minor differences in pharmacokinetic/pharmacodynamic parameters in children resulted in a faster time to onset and a shorter duration of neuromuscular block.

Older adult: The times to maximum block were approximately 1 minute slower in elderly patients.

Patient Counseling Points

(For additional information see "Cisatracurium: Patient drug information")

What is this drug used for?

- It is used to calm muscles during surgery.
- It is used to calm muscles while on a breathing machine.

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Seizures
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from

the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Nimbex | Sisacure;
- (AR) Argentina: Nimbium;
- (AT) Austria: Cisatracurium accord | Cisatracurium hikma | Cisatracurium hospira | Cisatracurium kalceks | Nimbex;
- (AU) Australia: Cisatracurium aft | Cisatracurium junio | Cisatracurium myx | Cisatracurium sxp | Dbl cisatracurium | Nimbex;
- (BD) Bangladesh: Nimbex;
- (BE) Belgium: Cisatracurium | Cisatracurium actavis | Cisatracurium Fresenius Kabi | Cisatracurium hospira | Cisatracurium kalceks | Cisatracurium sandoz | Nimbex;
- (BF) Burkina Faso: Nimbex;
- (BR) Brazil: Bedeci | Bes cisatracuri | Besilato de cisatracurio | Cis | Cisauni | Nimbium | Nosint | Volcis;
- (CI) Côte d'Ivoire: Nimbex;
- (CN) China: Cisatracurium besylate | Nimbex;
- (CO) Colombia: Alurium | Besilato de cisatracurio | Cirium | Cisatracurio | Cistrax | Kabicurium | Miocurim | Nimbium | Salatracurio | Tracurion;
- (CZ) Czech Republic: Cisatracurium hospira | Cisatracurium kabi | Cisatracurium kalceks | Nimbex;
- (DE) Germany: Cisatracurium | Cisatracurium accord | Cisatracurium actavis | Cisatracurium hexal | Cisatracurium kabi | Cisatracurium Ratiopharm | Cisatracurium Teva | Nimbex;
- (DO) Dominican Republic: Besilato de cisatracurio;
- (EC) Ecuador: Cis | Cistrac | Nimbex | Nimbium | Tracurion;
- (EE) Estonia: Cisatracurium norameda | Nimbex | Nimbex forte;
- (EG) Egypt: Nimbex;
- (ES) Spain: Cisatracurio Combino | Cisatracurio combino pharm | Cisatracurio Hospira | Cisatracurio inibsa | Cisatracurio kabi | Cisatracurio normon | Cisatracurio pfizer | Cisatracurio sala | Cisatracurio teva | Cisatracurium kalceks | Nimbex;
- (FI) Finland: Cisatracurium accord | Cisatracurium norameda | Nimbex;
- (FR) France: Cisatracurium | Cisatracurium accord | Cisatracurium hikma | Cisatracurium hospira | Cisatracurium kabi | Cisatracurium kalceks | Cisatracurium mylan | Nimbex;
- (GB) United Kingdom: Cisatracurium | Nimbex | Nimbex forte;
- (GR) Greece: Catren | Cisatracurium/kabi | Cisatracurium/teva | Cistrac | Nimbex;
- (HK) Hong Kong: Cisatracurium besylate | Cisatracurium kabi | Cisatracurium normon | Nimbex;
- (HU) Hungary: Cisatracurium actavis | Cisatracurium kabi | Cisatracurium kalceks | Nimbex;
- (IN) India: Cis atrabloc | Cistrac | Cistrac;
- (IT) Italy: Cisatracurio accord | Cisatracurio Actavis | Cisatracurio hikma | Cisatracurio kabi | Cisatracurio mylan | Cisatracurio sandoz | Cisatracurio teva | Nimbex;
- (JO) Jordan: Cistrac | Nimbex;
- (KE) Kenya: Cis curon | Nimbex;
- (KR) Korea, Republic of: Nimbex;
- (KW) Kuwait: Nimbex;
- (LB) Lebanon: Nimbex;
- (LT) Lithuania: Cisatracurium kabi | Cisatracurium kalceks | Cisatracurium norameda | Nimbex;
- (LU) Luxembourg: Nimbex;
- (LV) Latvia: Cisatracurium kabi | Cisatracurium kalceks | Cisatracurium norameda | Nimbex;
- (MX) Mexico: Bensitrak | Cicuryl | Cisatracurio | Nimbex;

- (MY) Malaysia: Cisatracurium kabi | Nimbex;
- (NL) Netherlands: Cisatracurium accord | Cisatracurium Fresenius Kabi | Nimbex;
- (NO) Norway: Cisatracurio accord | Cisatracurium accord | Cisatracurium kalceks | Nimbex;
- (PA) Panama: Nimbex;
- (PE) Peru: Nimbex;
- (PH) Philippines: Nimbex;
- (PK) Pakistan: Cis curon;
- (PL) Poland: Cisatracurium accord | Cisatracurium actavis | Cisatracurium hospira | Cisatracurium kabi | Cisatracurium kalceks | Cisatracurium mylan | Cisatracurium Noridem;
- (PR) Puerto Rico: Cisatracurium besylate | Nimbex;
- (PT) Portugal: Cisatracurio Combino | Cisatracurio Hospira | Cisatracurio kabi | Cisatracurio nor-mon;
- (QA) Qatar: Nimbex | Nimbex Forte;
- (RO) Romania: Nimbex;
- (RU) Russian Federation: Cisatracurii besilas | Nimbex;
- (SA) Saudi Arabia: Cis tac | Nimbex | Salocan | Sisacure | Traxia;
- (SE) Sweden: Nimbex;
- (SG) Singapore: Nimbex;
- (SI) Slovenia: Nimbex;
- (SK) Slovakia: Nimbex;
- (SR) Suriname: Nimbex;
- (SV) El Salvador: Cisatracurio;
- (TH) Thailand: Cisatracurium kabi | Nimbex;
- (TN) Tunisia: Cisatrex | Nimbex;
- (TR) Turkey: Nimbex;
- (TW) Taiwan: Cisatracurium kabi | Cisatral | Nimbex | Winbest;
- (UA) Ukraine: Cisatracurium kalceks;
- (UG) Uganda: Cis curon;
- (UY) Uruguay: Cis | Cisatracurio libra;
- (ZA) South Africa: Cisatracurium | Cisatracurium safeline | Cistrax | Nimbex;
- (ZM) Zambia: Nimbex;
- (ZW) Zimbabwe: Nimbex

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